RAG Assignment: Baseline vs. Re-Ranking

Document: *Dungeons & Dragons*

GitHub: https://github.com/SrushtiPekamwar/RAG

1. Architecture Comparison

A **bi-encoder** turns the question and each chunk into a vector on its own. The two never see each other, and the match is just the cosine similarity between the vectors. Because chunk vectors can be computed once and stored, search is very fast. The weakness is that a whole chunk is squeezed into one vector, so small but important words such as "not", "can't" or "only" can get lost.

A **cross-encoder** reads the question and the chunk together as one input. Self-attention lets every word of the question look at every word of the chunk, so it can tell whether the chunk really answers the question. The result is a more accurate score, but it has to be recomputed for every question–chunk pair.

2. Qualitative and Rank Shift Analysis

2.1 Rank changes

For each question I found the chunk(s) that contain the answer (the "gold" chunk) and tracked their best rank before and after re-ranking. The "Chunk" column gives the chunk number(s). For example, for the Cleave question chunk 67 moved from rank #6 to rank #1. A question counts as *rescued* when the gold chunk starts at rank 4–10 and ends in the top 3.

Type	Chunk	Question (short)	Before	After	Result
Exception	169, 170	Ritual spell uses a slot?	1	1	unchanged
Exception	47, 183	Teleporting and opportunity attacks	1	1	unchanged
Exception	122, 143	Advantage and disadvantage together	4	2	rescued
Negation	55	Healing above hit point maximum	1	3	still top 3
Negation	58, 59	Stable creature and death saves	1	1	unchanged
Negation	67	Cleave once per turn	6	1	rescued
Keyword trap	142	Cost of moving over rough ground	4	1	rescued
Buried chunk	26	Action for social checks (Influence)	8	3	rescued
Buried chunk	45, 134	Half cover and armor class	4	2	rescued
Re-ranking hurts	73, 124, 125	Penalty for untrained armor	4	9	worse
Re-ranking hurts	154	Damage of a makeshift object	4	7	worse

Type	Chunk	Question (short)	Before	After	Result
Control	58	Natural 20 on a death save	1	1	unchanged
Control	23, 24, 167	Second Reaction in a round	2	1	still top 3
Not in document	–	Price of the Player's Handbook	–	–	no answer chunk exists

Re-ranking rescued 5 of the 14 questions. Examples of single chunks: chunk 26 (Influence action) moved from rank 8 to 3, chunk 142 (Difficult Terrain) from 4 to 1, chunk 45 (cover table) from 4 to 2, and chunk 143 (advantage and disadvantage cancel) from 5 to 2.

2.2 Are the answers correct? Answer 1 vs. Answer 2

I checked every answer against the rulebook text myself. "No answer" means the model said the excerpts do not contain the answer, although the document does.

#	Question (short)	Answer 1	Answer 2
1	Ritual spell uses a slot?	Correct	Correct
2	Teleporting and opportunity attacks	Correct	Correct
3	Advantage and disadvantage together	Correct	Correct
4	Healing above hit point maximum	Correct	Correct
5	Stable creature and death saves	Correct	Correct
6	Cleave once per turn	No answer	Correct
7	Cost of moving over rough ground	Correct	Correct
8	Action for social checks (Influence)	No answer	Correct
9	Half cover and armor class	No answer	Correct
10	Penalty for untrained armor	No answer	No answer
11	Damage of a makeshift object	No answer	No answer
12	Natural 20 on a death save	Correct	Correct
13	Second Reaction in a round	Correct	Correct
14	Price of the Player's Handbook (not in document)	Correct	Correct
	Total	**9 correct, 5 no answer, 0 wrong**	**12 correct, 2 no answer, 0 wrong**

Hallucination, omission or misstatement? Neither pipeline hallucinated or stated a wrong rule. Every failure was an omission: the model left out the essential rule completely and said it could not find it. The baseline did this for questions 6, 8 and 9, where the critical chunk sat at rank 6, 8 and 4, outside its top 3. Re-ranking brought those chunks into the top 3, and Answer 2 then gave the right rule: "You can make this extra attack only once per turn" (Cleave), "The GM will typically ask you to take the Influence action", and "+2 bonus to AC" for half cover.

Why the baseline missed. For the cover question, the baseline's top 3 were chunks about armor training and armor categories, because they share the words "armor" and "class" with my question,

while the cover table (chunk 45) was at rank 4. The cross-encoder read the question as a whole and moved the cover table up.

Where re-ranking did not change the answer. For questions 3 and 7 the gold chunk moved up (rank 4 to 2, and 4 to 1), but the baseline already answered correctly, because other chunks in its top 3 also stated the rule. So a better rank does not always change the answer.

Where re-ranking made the ranking worse. For questions 10 and 11 the gold chunk moved down, from rank 4 to 9 and from 4 to 7, and for question 4 from 1 to 3. In question 4 the answer stayed correct. In questions 10 and 11 both pipelines said the answer was not in the excerpts, so they failed in the same way, and re-ranking did not create a new wrong answer. For question 10 the baseline's top 3 were chunks about proficiency bonuses and skills (they share the word "proficient"), and the re-ranked top 3 were no better. I think the cross-encoder was trained on web search passages, not game rules, so it can prefer text that merely discusses the same topic. This shows that re-ranking is not guaranteed to help, and that it can only reorder the 10 chunks the first stage found.

Overall. Across 14 questions, re-ranking turned 3 "no answer" replies into correct answers (questions 6, 8 and 9), left 2 failures unchanged (questions 10 and 11), and caused no new wrong answers. Answer 2 is also shorter, sometimes very short (for example just "+2 bonus to AC", which leaves out the Dexterity saving throw bonus).

2.3 When the answer is not in the document

For question 14 I asked how much the *Player's Handbook* costs in dollars, which the document never mentions. No chunk can answer it, so there is no gold chunk to track. Both pipelines retrieved unrelated chunks, and both answers said that the excerpts do not contain this information ("The provided rulebook excerpts do not contain information about the retail price of the D&D Player's Handbook in dollars"). I count both as correct, because saying "I don't know" is the right answer here, and nothing was made up. The baseline gave the same kind of reply in questions 6, 8 and 9, but there it was a failure, because the answer *was* in the document and had not been retrieved. So the same reply can be good or bad depending on whether the information exists. The prompt instruction "say so instead of guessing" is what stops the model from inventing a price. In the chatbot, this question is in the sidebar, and the answer is marked with an "oracle does not know" badge.

2.4 Failure modes covered

The tables above cover every failure mode the assignment lists:

- **Baseline stumbles, re-ranking restores relevance** (at least two needed, I have 5): questions 3, 6, 7, 8 and 9. The clearest are Cleave (chunk 67, rank 6 $\rightarrow$ 1) and the Influence action (chunk 26, rank 8 $\rightarrow$ 3).
- **Keyword trap:** question 7. I asked about moving over "rough ground", but the rulebook says "Difficult Terrain". The baseline's top 3 were chunks about Speed, burrowing and travel pace, which match my words about moving. The right chunk (142) moved from rank 4 to 1. The half cover question ("armor class" versus armor-training chunks) is a second example.

- **Exceptions and corner cases:** questions 1–3. A ritual spell does not expend a spell slot, teleporting does not provoke an opportunity attack, and advantage and disadvantage cancel each other. Only question 3 needed rescuing, because the baseline already ranked the others first.
- **Negation and constraints:** questions 4–6 (hit points "can't exceed" the maximum, a stable creature "doesn't make" death saves, an extra attack "only once per turn"). Question 6 was rescued from rank 6 to 1.
- **Other buried chunk:** questions 8 and 9, at ranks 8 and 4 before re-ranking.

I also included cases where re-ranking does not help: questions 10 and 11 (the gold chunk moved down), the easy control questions (no change), and question 14 (no answer exists in the document).

3. Latency Profiling

I timed retrieval only (not the LLM call, which is the same in both pipelines). Each of the 14 questions was run 5 times and the times were averaged. "Base" is Pipeline 1 (embed the question and compare with all 188 chunks). "CE" is the extra time the cross-encoder needs to score the 10 candidates. "Total" is Pipeline 2, which is Base plus CE, and "Slower" is Total divided by Base.

#	Question (short)	Base (ms)	CE (ms)	Total (ms)	Slower
1	Ritual spell uses a slot?	11.3	74.5	85.8	7.6×
2	Teleporting and opportunity attacks	8.1	71.9	80.0	9.9×
3	Advantage and disadvantage together	7.8	70.9	78.7	10.1×
4	Healing above hit point maximum	8.3	67.1	75.4	9.1×
5	Stable creature and death saves	8.6	64.1	72.8	8.5×
6	Cleave once per turn	9.6	97.0	106.5	11.1×
7	Cost of moving over rough ground	10.7	70.2	80.9	7.6×
8	Action for social checks (Influence)	9.2	73.5	82.7	9.0×
9	Half cover and armor class	7.8	65.4	73.2	9.4×
10	Penalty for untrained armor	8.9	60.9	69.8	7.8×
11	Damage of a makeshift object	7.7	68.8	76.5	9.9×
12	Natural 20 on a death save	7.8	71.3	79.1	10.1×
13	Second Reaction in a round	7.0	89.9	96.8	13.8×
14	Price of the Player's Handbook	9.1	68.1	77.2	8.5×
	Mean	**8.7**	**72.4**	**81.1**	**9.3×**

Re-ranking adds about 72 ms per question on average, which makes the full retrieval step roughly 9 times slower (between 7.6 and 13.8 times across the questions). In absolute terms this is small, because a single LLM call takes seconds, but the relative cost is real. The baseline step stayed between 7.0 and 11.3 ms for every question, while the cross-encoder step ranged from 60.9 to 97.0 ms. The cross-encoder was slowest for the Cleave (97.0 ms) and Reaction (89.9 ms) questions, probably because their candidate chunks were longer, although I did not measure that. The price

question, whose answer is not in the document, was in the normal range (68.1 ms for the cross-encoder).

4. Scaling Question

I do not run the cross-encoder on the whole document from the start because it has to read the question together with each chunk, so nothing can be prepared in advance. Every new question would need one full model run for every chunk. With 188 chunks that is fine, but the full SRD has about 1,500 chunks and real document collections have millions, so it would take far too long and cost too much. The bi-encoder avoids this: I embed every chunk once beforehand, and for a new question I only embed the question and compare vectors, which is very fast. So the sensible design is to use the cheap bi-encoder to pick a small shortlist (here 10 chunks) and then spend the expensive cross-encoder only on that shortlist. The catch is that the cross-encoder can only re-order what the first stage found, so a good chunk that is not in the top 10 can never be recovered.